

Module 2: Transport Layer and Network Layer



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Transport Layer

- Services and Protocols
- UDP and TCP

Hands-on

- Sockets Introduction
- Elementary TCP Sockets
- TCP Client/Server Example
- I/O Multiplexing
 - select()
 - poll()
- Elementary UDP Sockets
- Advanced I/O Functions

Network Layer

- Introduction
- Network-layer Protocols
- Unicast Routing
- Multicast Routing
- Multicasting Basics
- Intra-domain Routing
- Inter-domain Routing
- Next Generation IP (IPv6)
- Quality of Service (QoS)

Hands-on

- Linux Routing Table
- Routing Caches
- Routing Cache Implementation
- Adding Routes using ip

USER DATAGRAM PROTOCOL (UDP) — Introduction

- UDP (User Datagram Protocol) is a:
 - Connectionless protocol
 - Unreliable protocol
- Provides:
 - Process-to-process communication
- UDP adds very little functionality beyond IP.
- Main advantage:
 - Simplicity
 - Low overhead
 - High efficiency

USER DATAGRAM PROTOCOL (UDP) — Why Use UDP?

- UDP is useful when:
 - Messages are small
 - Reliability is not critical
 - Fast communication is required
- Requires less interaction than TCP.
- Suitable for applications needing quick request-response communication.

USER DATAGRAM PROTOCOL (UDP) — User Datagram Format

- UDP packet is called:

User Datagram

- Header size:



- Header consists of four fields:

- 1 Source Port Number
- 2 Destination Port Number
- 3 Length
- 4 Checksum

- Each field is:

16 bits = 2 bytes

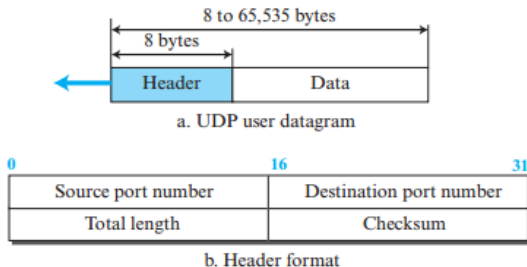
USER DATAGRAM PROTOCOL (UDP) — UDP Header Fields

Field	Size
Source Port Number	16 bits
Destination Port Number	16 bits
Length	16 bits
Checksum	16 bits

- Total header size = 8 bytes

User datagram packet format

Figure 3.39 *User datagram packet format*



USER DATAGRAM PROTOCOL (UDP) — Length Field

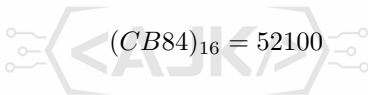
- Length field contains:
 - UDP Header + Data
- Maximum value: 
65535 bytes
- Actual size is smaller because UDP is encapsulated inside an IP datagram.

USER DATAGRAM PROTOCOL (UDP) — Example 3.11

UDP Header (Hexadecimal)

CB84 000D 001C XXXX

- Source Port:


$$(CB84)_{16} = 52100$$

- Destination Port:

$$(000D)_{16} = 13$$

- Total Length:

$$(001C)_{16} = 28 \text{ bytes}$$

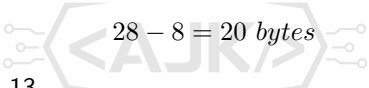
USER DATAGRAM PROTOCOL (UDP) — Example 3.11

Continued

- Header Length:

8 bytes

- Data Length:


$$28 - 8 = 20 \text{ bytes}$$

- Destination Port = 13
- Port 13 corresponds to:
 - Daytime Service
- Packet travels:
 - Client → Server

USER DATAGRAM PROTOCOL (UDP) — Process-to-Process Communication

- UDP provides:

Process-to-Process Communication

- Uses:
 - Socket Addresses
- Socket Address =

IP Address + Port Number

USER DATAGRAM PROTOCOL (UDP) — Flow Control

- UDP does not provide:
 - Flow Control
- No window mechanism exists.
- Receiver may be overwhelmed by incoming packets.
- If needed:
 - Application must implement flow control.

USER DATAGRAM PROTOCOL (UDP) — Error Control

- UDP does not provide:
 - Retransmission
 - Sequence Numbers
 - Acknowledgments
- Only error-detection mechanism:
 - Checksum
- Corrupted datagrams are silently discarded.

USER DATAGRAM PROTOCOL (UDP) — UDP Checksum

- Checksum is calculated using:

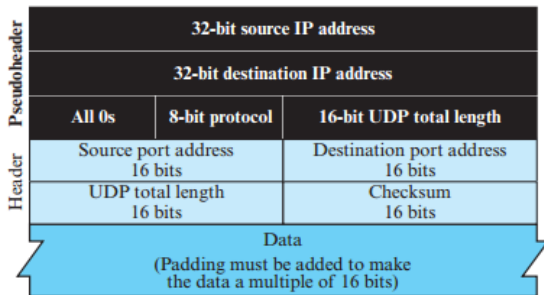
- 1 Pseudo Header
- 2 UDP Header
- 3 Data



- Used to detect transmission errors.
- Helps ensure datagram integrity.

Pseudoheader for checksum calculation

Figure 3.40 *Pseudoheader for checksum calculation*



USER DATAGRAM PROTOCOL (UDP) — Pseudo Header Purpose

- Pseudo header is not transmitted.
- Used only during checksum calculation.
- Detects corruption in:
 - Source IP
 - Destination IP
 - Protocol Field
- UDP Protocol Number:



USER DATAGRAM PROTOCOL (UDP) — Optional Checksum

- UDP checksum is optional.
- Checksum field = all 0s:
 - Checksum not calculated
- If computed checksum becomes all 0s:
 - Sender transmits all 1s

USER DATAGRAM PROTOCOL (UDP) — Connectionless Service

- Every UDP datagram is independent.
- No:
 - Connection Establishment
 - Connection Termination
- Datagrams:
 - May follow different paths
- No relationship between packets.

USER DATAGRAM PROTOCOL (UDP) — Congestion Control

- UDP does not provide:
 - Congestion Control
- Assumes:
 - Small
 - Sporadic traffic
- Suitable for many real-time applications.

USER DATAGRAM PROTOCOL (UDP) — Queuing

- UDP associates queues with ports.
- Client process requests:
 - Port Number
- Operating system creates:
 - Incoming Queue
 - (Sometimes) Outgoing Queue

USER DATAGRAM PROTOCOL (UDP) — Multiplexing and Demultiplexing

- One UDP module serves multiple applications.
- Multiplexing:
 - Multiple processes share UDP
- Demultiplexing:
 - Incoming datagrams delivered to correct process

USER DATAGRAM PROTOCOL (UDP) — UDP vs Simple Protocol

- UDP is similar to:
 - Generic Connectionless Simple Protocol
- Difference:
 - Optional Checksum
 - No feedback to sender.
 - Corrupted packets are discarded.

USER DATAGRAM PROTOCOL (UDP) — Advantages of UDP

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- A large, light gray watermark logo is centered in the background of the list. It features the letters 'AJK' in a bold, sans-serif font, enclosed within a stylized hexagonal frame with circuit-like lines extending from the corners.
- ① Simple Protocol
 - ② Low Overhead
 - ③ Fast Communication
 - ④ No Connection Setup Delay
 - ⑤ Supports Multicasting
 - ⑥ Suitable for Real-Time Applications

USER DATAGRAM PROTOCOL (UDP) — Typical Applications

- DNS
- TFTP
- SNMP
- RIP
- Multicasting
- VoIP
- Video Streaming
- Online Gaming
- Interactive Real-Time Applications



USER DATAGRAM PROTOCOL (UDP) — When to Use UDP?

Requirement	UDP Suitable?
Short Request-Response	Yes
Real-Time Audio/Video	Yes
Multicasting	Yes
Large File Transfer	No
Reliable Delivery Required	No